Low Complexity Complex Matrix Inversion Method for MIMO Communication Systems

Denggao Yu*, Zhaohua Lu‡, Shiwen He*†, Bo Wu*, Yongming Huang*, Luxi Yang*
*School of Information Science and Engineering, Southeast University, Nanjing 210096, China.
†The State Key Laboratory of Millimeter Waves, Department of Radio Engineering,
Southeast University, Nanjing 210096, China.
Email: {220130708, shiwenhe, 220130654, huangym, lxyang}@seu.edu.cn
‡Zhongxing Telecom Equipment, Shenzhen, 518000, China. Email: lu.zhaohua@zte.com.cn

***Abstract*—In multiple-input multiple-output (MIMO) communication systems, complex matrix inversion is a very computationally demanding operation. Especially when the number of antennas increases, i.e., in a massive MIMO system, the complexity of matrix inversion becomes very high. Motivated by this observation, a new low complexity complex matrix inversion method, called SDF-SGR (Square root and Division Free Squared Givens' Rotations) based algorithm, is designed for MIMO channels. Square root operation is avoided in the whole algorithm, and division operation is replaced by shift operation during the Givens' rotations phase. Besides, since the scale factor z, involved in the traditional SGR, has little influence on the whole algorithm, it is removed in this SDF-SGR algorithm. Considering a 4×4 complex matrix inversion, the SDF-SGR based algorithm could reduce the multiplication operations by 13.07% compared with the traditional SGR algorithm, and the division operation is reduced by almost 52.94%.**

***Index Terms*—matrix inversion, low complexity, multiple-input multiple-output, givens' rotations**

I. Introduction

Multiple-input multiple-output (MIMO) systems equipped with multiple antennas at both the transmitting and receiving sides of the radio link, have been utilized in many communication systems, such as IEEE 802.11ac, IEEE 802.11n, and 3GPP-LTE, to greatly enhance the performance of wireless communications [1–4]. In order to fully exploit the potential advantage provided by MIMO systems, many advanced transceiving techniques are usually adopted, such as space-time coding, spatial multiplexing, beamforming and etc [5]. However, most of these algorithms require complex matrix inversion, e.g. the zero-forcing (ZF) and minimum mean-squared error (MMSE) decoding [6]. When the number of antennas increases, the matrix inversion is hard to be implemented in hardware because of its high complexity [7]. This has become a serious obstacle to implement massive MIMO. Consequently, it is necessary to figure out an efficient matrix inversion algorithm with low complexity.

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Matrix inversion is an essential operation in many signal processing or numerical problems, such as digital filtering [8], 3D rendering [9], signal detection [6], etc. And it is well elaborated in previous publications [10]. Strassen [11], Strassen-Newton [12], Gaussian elimination [13], Gauss-Jordan [13], QR decomposition [14], Cholesky decomposition [15], etc, are some of the common used algorithms for solving the matrix inversion problems. For large matrix, QR decomposition is traditionally applied to the original matrix to generate a unitary matrix $\mathbf{Q}$ and an upper triangular matrix $\mathbf{R}$, then back substitution could be used to compute the inverse matrix of $\mathbf{R}$. And the inverse matrix of $\mathbf{Q}$ is simply its Hermitian transpose. QR decomposition is a method that is widely studied for large matrix inversion, such as Gram-Schmidt transform, Householder matrices, and Givens' rotations [10]. But the disadvantage of these algorithms is that they require high-complexity square root operations. Recently, Squared Givens' Rotations (SGR) [16] has attracted much attention for hardware implementation. The SGR algorithm could avoid the square root operations and reduce the number of multiplications by half [17,18]. However, the SGR algorithm could hardly handle the situations when zeros occur on the diagonal elements of the matrix either initially or during processing. In [19], a modified SGR (MSGR) algorithm was proposed to overcome this drawback. But many division operations are still involved in the MSGR algorithm, which may be further reduced.

In this paper, we focus on designing a method to further reduce the complexity of matrix inversion. The simulation results show that, compared with the traditional SGR algorithm, the SDF-SGR based algorithm could reduce the multiplication and division operations by 13.07% and 52.94% respectively, when a 4×4 complex matrix inversion is considered. The remainder of this paper is organized as follows. In section II, the MIMO communication system is briefly introduced. In section III, the low complexity CMI (Complex Matrix Inversion) method is presented. In section IV, we analyze the complexity of the whole algorithm. And the simulation results are illustrated in section V. Then we finally conclude this paper in Section VI.

II. System Model

Considering the complex baseband model for MIMO channel with M transmitting antennas and N receiving antennas,

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as shown in Fig. 1, the relation between the transmitted signal and the received signal is formulated as

$$\mathbf{r} = \mathbf{HFs} + \mathbf{n}, \tag{1}$$

where $\mathbf{r} \in C^{N\times 1}$ is the received signal, $\mathbf{s} \in C^{M\times 1}$ is the transmitted signal, $\mathbf{H} \in C^{N\times M}$ is the channel matrix, $\mathbf{F}$ is the precoding matrix, and $\mathbf{n} \in C^{N\times 1}$ represents the white Gaussian noise.

In this paper, we treat interference as noise and consider linear receiving beamforming strategy, then the estimated signal is expressed as: $\hat{\mathbf{s}} = \mathbf{Wr}$, where $\mathbf{W}$ is the equalizer such as ZF or MMSE as given by

$$\mathbf{W}_{ZF} = \left(\mathbf{H}_E^{\dagger}\mathbf{H}_E\right)^{-1}\mathbf{H}_E^{\dagger} \tag{2}$$

$$\mathbf{W}_{MMSE} = \left(\mathbf{H}_E^{\dagger}\mathbf{H}_E + \sigma_n^2\mathbf{I}_M\right)^{-1}\mathbf{H}_E^{\dagger}, \tag{3}$$

where $(\cdot)^{\dagger}$ denotes conjugate-transpose operation, $\mathbf{H}_E$ is the equivalent channel matrix, i.e., $\mathbf{H}_E = \mathbf{HF}$, σ_n^2 is the average noise power and $\mathbf{I}_M$ is an $M \times M$ identity matrix. Note that in equations (2) and (3), both $\mathbf{W}_{ZF}$ and $\mathbf{W}_{MMSE}$ need complex matrix inversion. It should be pointed out that the matrix inversion operation may also be used in the computation of the precoding matrix $\mathbf{F}$. In a MIMO system with increasing number of antennas, high dimension matrix inversion is usually needed. Motivated by these observations, we focus on designing a low complexity algorithm to address the matrix inversion problem in the following description.

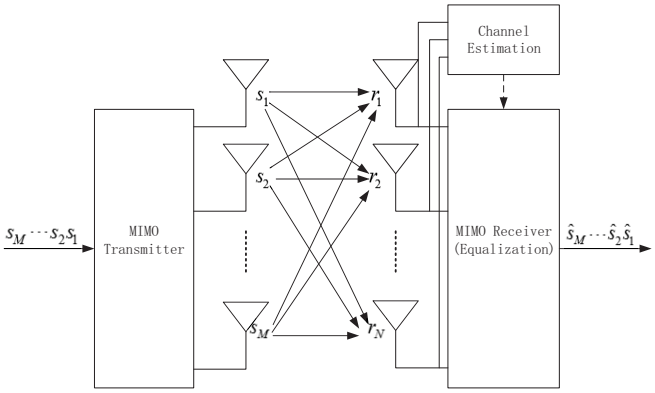


Fig. 1: M × N MIMO system model

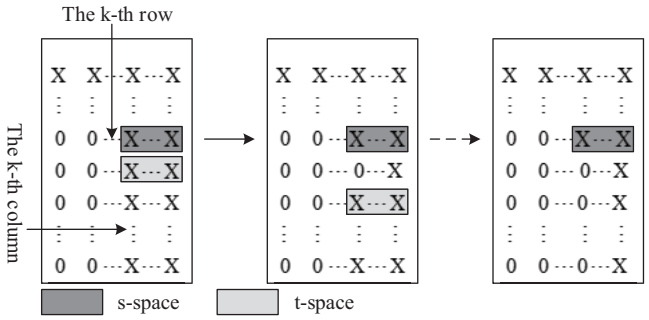


Fig. 2: Elements elemination

III. Low Complexity CMI Method Design

By using QR decomposition (QRD), matrix $\mathbf{M}$ can be decomposed into a unitary matrix $\mathbf{Q}$ and an upper triangular matrix $\mathbf{R}$, namely, $\mathbf{M} = \mathbf{QR}$ [10]. And it has been proved that QR decomposition could be also written as [16]

$$\mathbf{M} = \mathbf{M}_A\mathbf{M}_U^{-1}\mathbf{N}, \tag{4}$$

where, $\mathbf{M}_A = \mathbf{QM}_R$, $\mathbf{M}_R = diag(\mathbf{R})$, $\mathbf{M}_U = diag(\mathbf{N}) = \mathbf{M}_R^2$, and $\mathbf{N}$ is an upper triangular matrix, and $\mathbf{N} = \mathbf{M}_R\mathbf{R}$, $diag(\mathbf{R})$ returns the main diagonal of matrix $\mathbf{R}$. Then the inversion of matrix $\mathbf{M}$ is calculated as,

$$\mathbf{M^{-1}} = \mathbf{N^{-1}}\left(\mathbf{M_A M_U^{-1}}\right)^{-1}. \tag{5}$$

In subsection III-A, the SDF-SGR based algorithm will be introduced to show how the upper triangular matrix $\mathbf{N}$ could be obtained. And the back substitution method, which is usually used to calculate the inversion of a triangular matrix, will be described in subsection III-B. The calculation of $\left(\mathbf{M_A M_U^{-1}}\right)^{-1}$ will be presented in subsection III-C.

A. SDF-SGR based algorithm

Let $\mathbf{a}$ and $\mathbf{b}$ be two complex vectors, as shown as follows,

$$\begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & \cdots & a_k & \cdots & a_p \\ b_1 & b_2 & \cdots & b_k & \cdots & b_p \end{bmatrix}. \tag{6}$$

Assume that $a_k = 0$ and $b_k = 0$, then the traditional Givens' rotation could be done to eliminate b_k in vector $\mathbf{b}$.

$$\begin{cases} c = \left(a_k^* a_k + b_k^* b_k\right)^{1/2} \\ \bar{\mathbf{a}} = c^{-1}\left(a_k^*\mathbf{a} + b_k^*\mathbf{b}\right) \\ \bar{\mathbf{b}} = c^{-1}\left(-b_k\mathbf{a} + a_k\mathbf{b}\right), \end{cases} \tag{7}$$

where $\bar{\mathbf{a}}$ and $\bar{\mathbf{b}}$ are the updated vectors of $\mathbf{a}$ and $\mathbf{b}$. It could be seen that, in the updated vector $\bar{\mathbf{b}}$, the kth element $\bar{b}_k = 0$. Meanwhile square root operations are required in equation (7), which cause the whole complexity high. To overcome this, the SGR [16] propose a method as follows.

Vectors $\mathbf{a}$ and $\mathbf{b}$ are translated firstly to $\mathbf{s}$ and $\mathbf{t}$ -space respectively, i.e.,

$$\begin{cases} \mathbf{s} = a_k^*\mathbf{a} \\ \mathbf{t} = \mathbf{b}z^{-(1/2)}, \end{cases} \tag{8}$$

where $(\cdot)^*$ is the complex conjugate operation. And $z > 0$ is a scale factor. Then, the Givens' rotation equation (7) could be replaced as,

$$\begin{cases} \bar{\mathbf{s}} = \mathbf{s} + zt_k^*\mathbf{t} \\ \bar{\mathbf{t}} = \mathbf{t} - \frac{t_k}{s_k}\mathbf{s} \\ \bar{z} = z\frac{s_k}{\bar{s}_k}, \end{cases} \tag{9}$$

As could be seen in (9), The kth element in vector $\bar{\mathbf{t}}$ satisfies $\bar{t}_k = 0$. And the other elements under the kth row in the kth column could be eliminated sequently by rotating $\mathbf{M}$ as indicated in Fig. 2. Then the upper triangular matrix $\mathbf{N}$ could be obtained when other columns are implemented the similar operations.

To further reduce the complexity of the Givens' rotations, [19] removed the scale factor z from the equations above by the observation that the scale factor could not affect the zero properties of the vectors. Then the operations related to the scale factor z can be saved, which will definitely contribute to the complexity reducing. And many simulations have been done to prove that the method without the scale factor z is

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more or less the same accurate as the comparable techniques. Thus we have,

$$\begin{cases} \bar{\mathbf{s}} = \mathbf{s} + t_k^* \mathbf{t} \\ \bar{\mathbf{t}} = \mathbf{t} - \frac{t_k}{s_k}\mathbf{s}. \end{cases} \tag{10}$$

Rewrite the second equation in (10) as $\bar{\mathbf{t}} = (s_k\mathbf{t} - t_k\mathbf{s})/u_k$. From equation (8), it could be seen that s_k is a non negative real number, here we firstly consider s_k to be a positive real number, which has little effect on the properties of the vector $\bar{\mathbf{t}}$. Then the division operation could be replaced by $\bar{\mathbf{t}} = (s_k\mathbf{t} - t_k\mathbf{s})/s'_k$, here s'_k is the closest number to s_k, which satisfies $s'_k{=}2^x$, where x is an integer. Take $s_k = 10.31$ as an example, then s'_k could be 8. Since s'_k is power of 2, the division operations then could be done by just shifting the numerators. Thus, equation (10) could be expressed as

$$\left.\begin{array}{l} \bar{\mathbf{s}} = \mathbf{s} + t_k^* \mathbf{t} \\ \bar{\mathbf{t}} = s_k\mathbf{t} - t_k\mathbf{s} \\ \exp(\bar{\mathbf{t}}) = \exp(\bar{\mathbf{t}}) - \exp(s'_k) \end{array}\right\} when \;\; s_k \neq 0, \tag{11}$$

where $\exp(r)$ denotes the exponent of the number r. Apparently, division and squared root operations are not required in (11).

Then we should consider the situations when s_k equal to zero, which may actually occur either during the rotations or in the original matrix. With $s_k = 0$, the equation (11) may implement wrong operations. Fortunately, a simple solution represented as follows could effectively solve this problem.

$$\left.\begin{array}{l} \bar{\mathbf{s}} = \mathbf{t} \\ \bar{\mathbf{t}} = -\mathbf{s} \end{array}\right\} when \;\; s_k = 0. \tag{12}$$

Then the whole SDF-SGR method is comprised by equations (11) and (12), in which division and square root operations are totaly removed.

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_1 \\ \mathbf{M}_2 \\ \vdots \\ \mathbf{M}_n \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1n} \\ M_{21} & M_{22} & \cdots & M_{2n} \\ \vdots & \vdots & \vdots & \\ M_{n1} & M_{n2} & \cdots & M_{nn} \end{bmatrix}. \tag{13}$$

Considering a $n \times n$ complex matrix $\mathbf{M}$ as given in (13), the following pseudo codes ,i.e., Algorithm 1, will show how the upper triangular matrix $\mathbf{N}$ is obtained from $\mathbf{M}$ by SDF-SGR method.

B. Back substitution Method

The upper triangular matrix $\mathbf{N}$ could be calculated by the SDF-SGR algorithm via rotating the original matrix $\mathbf{M}$ as described in subsection-A. From equation (5), we know that $\mathbf{N}^{-1}$ should be firstly obtained before getting the inversion of matrix $\mathbf{M}$. As known to us all, back substitution method is usually used to achieve the inversion of the triangular matrix [10], i.e.,

$$\mathbf{G}_{ij} = \begin{cases} -\frac{1}{\mathbf{N}_{jj}} \left(\sum_{k=i}^{j-1} \mathbf{G}_{ik}\mathbf{N}_{kj} \right) & i < j \\ \frac{1}{\mathbf{N}_{jj}} & i = j \\ 0 & i > j \end{cases} \tag{14}$$

here $\mathbf{G} = \mathbf{N}^{-1}$.

Algorithm 1 SDF-SGR based algorithm

```
n ← dimension of matrix M
s_i ← M_i ← the i-th row of matrix M(i = 1, 2, ··· , n)
for k ← 1 To n
    if s_kk ≠ 0 then
        s_k ← s*_kk s_k
    else
        s_k ← s_k
    end if
    for j ← k + 1 To n
        t_j ← s_j
        if s_kk ≠ 0 then
        compute s̄_k, t̄_j by Equation (11)
        else
          compute s̄_k, t̄_j by Equation (12)
        end if
        M_k ← s_k ← s̄_k
        M_j ← s_j ← s̄_j
    end for
end for
```

C. Matrix Inversion

In this subsection, we will introduce how to calculate $\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$, since $\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$ should also be achieved before getting the $\mathbf{M}^{-1}$. After the upper triangular matrix $\mathbf{N}$ is achieved, $\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$ could be calculated as follows.

Since the equation (4) could be written as $\mathbf{N} = \left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}\mathbf{M}$, then $\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$ could be treated as a whole factor ψ. The upper triangular matrix $\mathbf{N}$ could be produced from $\mathbf{M}$ via left multiplied by ψ. Then we could conclude that the $\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$ could be obtained when the identity matrix $\mathbf{I}$ is left multiplied by ψ, i.e., $\left(\mathbf{Q}_A\mathbf{D}_U^{-1}\right)^{-1} = \psi\mathbf{I}$, which means $\mathbf{I}$ could be rotated in the similar way as $\mathbf{M}$ is rotated in subsection A to generate $\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$. Recalling subsection B, $\mathbf{N}^{-1}$ is available by back substitution. And $\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$ could be calculated in this subsection. Then the final inversion matrix of $\mathbf{M}$ could be obtained as $\mathbf{M}^{-1} = \mathbf{N}^{-1}\left(\mathbf{M}_A\mathbf{M}_U^{-1}\right)^{-1}$.

IV. COMPLEXITY ANALYSIS

In this section, we mainly analyze the complexity of the SDF-SGR based algorithm described above. Table I shows the complexity of SDF-SGR based algorithm and several traditional matrix inversion algorithms in MIMO communication systems composed of n_t transmitting antennas and n_r receiving antennas ($n_t = n_r$). In Table I, the numbers behind the expressions show an absolute complexity measure for $n_r = 4$.

As Table I shows, all the algorithms exhibit $O\left(n_r^3\right)$ additions and multiplications. But Cholesky Decomposition and Conventional Givens' Rotations QRD suffer from square root operations. Furthermore, Cholesky Decomposition has a limitation that the matrix should be positive definite.

Compared with Squared Givens' Rotations algorithm, the SDF-SGR based algorithm suffers less division operations,

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TABLE I: Matrix Inversion Complexity comparison

Algorithm	Additions($n_r = 4$)	Multiplications($n_r = 4$)	Square roots($n_r = 4$)	Divisions($n_r = 4$)
Cholesky Decomposiition [10]	$2n_r^3 - n_r^2 + n_r$ (116)	$2\left(n_r^3 + n_r\right)$ (136)	n_r (4)	$2n_r^2 - n_r$ (28)
Gaussian [10]	$4n_r^3 - n_r^2 - 3n_r$ (228)	$4n_r^3 + 2n_r^2 - 6n_r$ (264)	0 (0)	$2\left(n_r^2 - n_r\right)$ (24)
Square Root Decomposition [20]	$5n_r^3 + 3n_r^2$ (368)	$9n_r^3 + 7n_r^2$ (688)	0 (0)	n_r^2 (16)
Conventional Givens' Rotations QRD [10]	$8n_r^3 + n_r^2 - 6n_r$ (504)	$\frac{32}{3}n_r^3 + 11n_r^2 - \frac{47}{3}n_r$ (796)	$\frac{3}{2}\left(n_r^2 - n_r\right)$ (18)	$3n_r^2 - n_r$ (44)
Squared Givens' Rotations QRD [16]	$8n_r^3 - 4n_r^2$ (448)	$\frac{28}{3}n_r^3 + \frac{7}{2}n_r^2 + \frac{7}{6}n_r$ (658)	0 (0)	$\frac{5}{2}n_r^2 - \frac{3}{2}n_r$ (34)
SDF-SGR based algorithm	$\frac{25}{3}n_r^3 - 4n_r^2 - \frac{1}{3}n_r$ (468)	$8n_r^3 + 3n_r^2 + 3n_r$ (572)	0 (0)	n_r^2 (16)

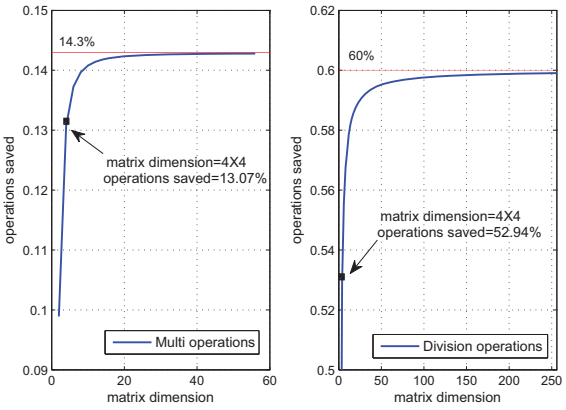


Fig. 3: Operations Saved Analysis of SDF-SGR based algorithm

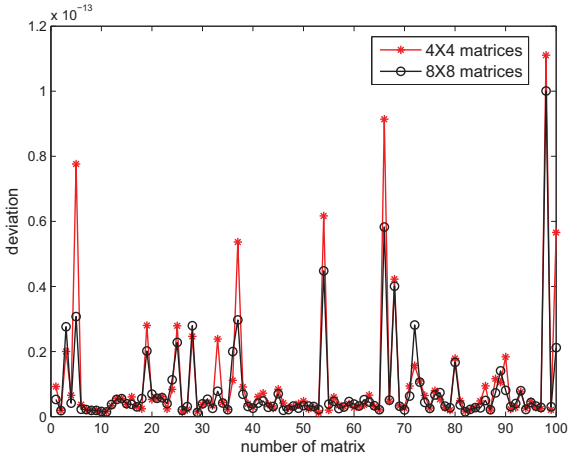


Fig. 4: Accuracy Analysis of SDF-SGR based algorithm

which are involved in the back substitution part barely. Additionally, SDF-SGR based algorithm has less multiplications because the scale factor z is removed. And the increased additions are acceptable, which are introduced by the shift operations. In Fig. 3, we analysis the multiplication and division operations saved by SDF-SGR based algorithm compared with Squared Givens' Rotations. The x axis denotes the dimension of the matrix, and the y axis denotes the percentage of operations saved. It could be seen that, with the increase of the matrix dimension, the multiplication operations could be saved by almost 14.3%, and the division operations could be saved by up to 60%. And when $n_r = 4$, the division and multiplication operations could be reduced by 52.94% and 13.07% respectively.

V. Simulation Results

In this section, we will verify the accuracy of the SDF-SGR based algorithm and assess its fixed-point performance.

In Fig. 4, one hundred 4×4 and 8×8 MIMO Rayleigh-fading channel matrices are generated randomly. Then the inversion of each matrix is computed by the SDF-SGR based algorithm. We compare the inversion matrices with the results applied by NAG (Numerical Algorithms Group) Matlab toolkit. The x axis denotes the number of matrix. The y axis enumerates the deviation error. From the figure, the SDF-SGR based algorithm achieved almost the same result as the NAG Matlab toolkit, the deviation is less than 10^{-13}. According to Fig. 4, we could conclude that the accuracy of SDF-SGR based algorithm is guaranteed.

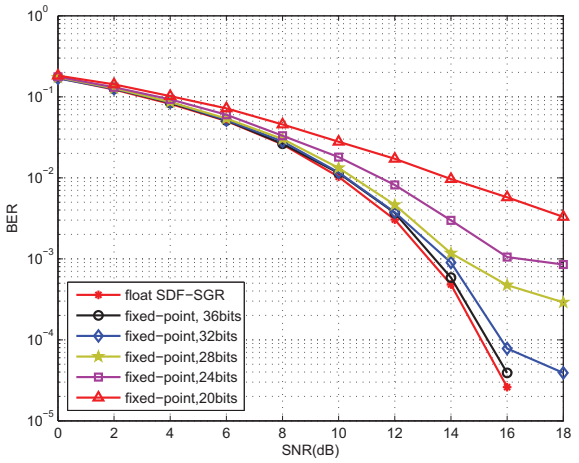


Fig. 5: Fixed-Point BER performance of SDF-SGR based MIMO transceiver

The fixed point performance of SDF-SGR based algorithm is illustrated in Fig. 5. The y axis is the BER performance of a 4×4 MIMO receiver on a Rayleigh-fading channel. For 36 bits fixed-point, the BER performance is almost the same as the float SDF-SGR based receiver. For 32 bits fixed-point, there are some degradations in terms of BER performance at high SNR. And for less bits fixed-point, the BER performance is significantly decreased in this simulation.

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VI. Conclusion

In this paper, a low complexity complex matrix inversion method was presented for MIMO channels. During the whole algorithm, square root operations were avoided, and division operations were replaced by shift operations during the Givens' rotations phase. Besides, the scale factor z was also removed, which could help reduce the multiplications. The simulation results have shown that, with the increase of the matrix dimension, the reduced multiplication and division operations could be up to 14.3% and 60% respectively. In a MIMO communication system, in which $N_r = N_t = 4$, the multiplications are reduced by 13.07%, and the division operations are reduced by 52.94%. According to these, we could conclude that, the SDF-SGR based algorithm could effectively reduce the complexity of the complex matrix inversion problem. The fixed-point and BER performance of this algorithm are also simulated, which illustrated that the acceptable fixed-point performance could be achieved with suitable bits.

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